Townsend P9.11

$$\frac{N_\ell}{N_0} = \underbrace{(2\ell+1)}\, e^{-(E_\ell - E_0)/k_B T}$$

↳ degeneracy of the level with angular momentum $L^2 = \ell(\ell+1)\hbar^2$

(9.107) $$E_\ell = \frac{\ell(\ell+1)\hbar^2}{2I}$$

(a) $$\frac{N_\ell}{N_0} = (2\ell+1)\, e^{-\frac{\hbar^2 \ell(\ell+1)}{2 I k_B T}}$$

Find maximum: $$\frac{dN_\ell}{d\ell} = 0 \Rightarrow 2 e^{-\frac{\hbar^2 \ell(\ell+1)}{2 I k_B T}} - (2\ell+1)\cdot\frac{\hbar^2}{2 I k_B T}\cdot \frac{d}{d\ell}(\ell^2+\ell)\, e^{-\frac{\hbar^2 \ell(\ell+1)}{k_B T}}$$

$$\Rightarrow 2 - (2\ell+1)^2 \frac{\hbar^2}{2 I k_B T} = 0$$

$$\Rightarrow (2\ell+1)^2 = \frac{4 I k_B T}{\hbar^2} \Rightarrow 2\ell+1 = \sqrt{\frac{4 I k_B T}{\hbar^2}}$$

$$\Rightarrow \ell_m = \frac{\sqrt{I k_B T}}{\hbar} - \frac{1}{2}$$

$$\frac{N_{\ell_m}}{N_0} = \sqrt{\frac{4 I k_B T}{\hbar^2}}\; e^{-\frac{\hbar^2}{2 I k_B T}\left(\frac{\sqrt{I k_B T}}{\hbar} - \frac{1}{2}\right)\left(\frac{\sqrt{I k_B T}}{\hbar} + \frac{1}{2}\right)}$$

$$= \sqrt{\frac{4 I k_B T}{\hbar^2}}\; e^{-\frac{1}{2} + \frac{\hbar^2}{8 I k_B T}}$$

(b) At room $T \approx 300K \Rightarrow k_B T \approx \frac{1}{40}$ eV

$I = \mu r_0^2 = \frac{\lambda \ell \hbar}{2\pi c}$ from Eq. (9.110)

We need $\frac{I}{\hbar^2} = \frac{\lambda l}{2\pi\hbar c} = \frac{0.0486\ \text{cm}}{6.63\cdot 10^{-34}\ \text{J}\cdot\text{s}\cdot 10^{8}\ \text{m/s}} =$

$$= \frac{4.86\cdot 10^{-4}\ \text{m}}{6.63\cdot 10^{-26}\ \text{J}\cdot\text{m}} = 0.7\cdot 10^{22}\ \text{J}^{-1}$$

Thus $\frac{I}{\hbar^2} k_B T = 0.7\cdot 10^{22}\ \frac{1}{\text{J}}\cdot\frac{1.6\cdot 10^{-19}\ \text{J}}{40} = \frac{1.2\cdot 10^{3}}{40} = 30$

$\boxed{\text{Note: } 1\,\text{eV} = 1.6\cdot 10^{-19}\ \text{J}}$

~~That~~ Thus $l_{max} = \sqrt{30} - \frac{1}{2} \approx 4.5 \approx 4$

roughly consistent with absorption with $l \simeq 4$ on Figure 9.7

Griffiths <u>P5.28</u> Evaluate at $T=0$

$$N = \frac{V}{2\pi^2}\int_0^\infty \frac{k^2}{e^{\left(\frac{\hbar^2 k^2}{2m}-\mu\right)/k_BT}+1}\,dk$$

$$E = \frac{V}{2\pi^2}\frac{\hbar^2}{2m}\int_0^\infty \frac{k^4}{e^{\left(\frac{\hbar^2 k^2}{2m}-\mu\right)/k_BT}+1}\,dk$$

At $T=0$ $\mu = E_F$, see Fig. 5.8

Thus $\frac{1}{e^{\left(\frac{\hbar^2 k^2}{2m}-\mu\right)/k_BT}+1} \to \begin{cases} 1, & \frac{\hbar^2 k^2}{2m} - E_F \le 0 \\ 0, & \frac{\hbar^2 k^2}{2m} - E_F > 0 \end{cases}$

$\Rightarrow$ the integral $\int_0^\infty \to \int_0^{k_F}$ where

$k_F = \sqrt{\frac{2mE_F}{\hbar^2}}$ is the Fermi momentum

(1) $$N = \frac{V}{2\pi^2}\int_0^{k_F} k^2\,dk = \frac{V}{2\pi^2}\,\frac{k_F^3}{3} = \frac{V}{6\pi^2}\left(\frac{2mE_F}{\hbar^2}\right)^{3/2}$$

(2) $$E = \frac{V}{2\pi^2}\,\frac{\hbar^2}{2m}\int_0^{k_F} k^4\,dk = \frac{V}{2\pi^2}\,\frac{\hbar^2}{2m}\,\frac{k_F^5}{5}$$

To account for $s = 1/2$ of the electrons, we must multiply right-hand sides of (1) and (2) by $\boxed{g = 2s+1}$, which represents <u>spin degeneracy</u> (which is NOT accounted for in eq. 5.97!)

Thus (1) $$N = \frac{V}{3\pi^2}\left(\frac{2mE_F}{\hbar^2}\right)^{3/2}$$

$$\Rightarrow \quad E_F = \frac{\hbar^2}{2m}\left(3\pi^2\frac{N}{V}\right)^{2/3} \quad \text{as in (5.43)}$$

and (2) $$E = \frac{V}{\pi^2}\,\frac{\hbar^2}{2m}\,\frac{k_F^5}{5}\cancel{\left(\frac{2mE_F}{\hbar^2}\right)} = \frac{\hbar^2 k_F^5 V}{10\pi^2 m} \quad \text{as in (5.45)}$$

P5.30 (a) $$\rho(\omega) = \frac{\hbar\omega^3}{\pi^2c^3\left(e^{\hbar\omega/k_BT}-1\right)}$$

$$k = \frac{2\pi}{\lambda} = \frac{\omega}{c} \quad \Rightarrow \begin{cases} \lambda = \dfrac{2\pi c}{\omega} \\ \omega = \dfrac{2\pi c}{\lambda} \end{cases} \Rightarrow d\omega = -\frac{2\pi c}{\lambda^2}\,d\lambda$$

just means that increasing ω implies decreasing λ

Clearly $\rho(\omega)\,d\omega = \rho(\lambda)\,d\lambda$

$$\Rightarrow \rho(\omega)\,d\omega = \frac{\hbar\left(\frac{2\pi c}{\lambda}\right)^3}{\pi^2c^3\left(e^{\frac{\hbar\cdot 2\pi c}{k_BT\lambda}}-1\right)}\cdot\frac{2\pi c}{\lambda^2}\,d\lambda =$$

$$= \frac{\hbar\cdot(2\pi c)^4\,d\lambda}{\pi^2c^3\lambda^5\left(e^{hc/\lambda k_BT}-1\right)}$$

$$\Rightarrow \quad \frac{16\hbar\pi^2 c \; d\lambda}{\lambda^5\left(e^{\frac{hc}{\lambda k_B T}}-1\right)} = \rho(\lambda)\, d\lambda$$

$$\Rightarrow \rho(\lambda) = \frac{16\pi^2 \hbar c}{\lambda^5\left(e^{\frac{hc}{\lambda k_B T}}-1\right)}$$

where we use $h = 2\pi\hbar$

(b) To Find λ_{max} at which $\rho(\lambda_{max})$ is maximized, set

$$\frac{d\rho(\lambda)}{d\lambda} = 0$$

$$\Rightarrow 0 = 16\pi^2\hbar c \left\{ \frac{-5}{\lambda^6\left(e^{\frac{hc}{\lambda k_B T}}-1\right)} - \frac{e^{\frac{hc}{\lambda k_B T}} \frac{hc}{k_B T}\left(-\frac{1}{\lambda^2}\right)}{\lambda^5\left(e^{\frac{hc}{\lambda k_B T}}-1\right)^2} \right\}$$

$$\Rightarrow \quad 5\left(e^{\frac{hc}{\lambda k_B T}}-1\right) = \frac{hc}{\lambda k_B T}\, e^{\frac{hc}{\lambda k_B T}}$$

let $x = \frac{hc}{\lambda k_B T}$ $\Rightarrow$ $5(e^x - 1) = x e^x$

$$5\left(1 - e^{-x}\right) = x$$

$$5 - x = 5e^{-x}$$

Mathematica finds $x = 4.966$

$$\Rightarrow \lambda_{max} = \frac{hc}{(4.966)\, k_B T} = \frac{2.9 \cdot 10^{-3}\, m \cdot K}{T}$$

Kelvin

this is Wien displacement law.

P5.31 The energy density $N_\omega \hbar\omega / V = \rho(\omega)\, d\omega$ from (5.113)

$$\rho(\omega) = \frac{\hbar \omega^3}{\pi^2 c^3 \left(e^{\hbar\omega/k_B T} - 1\right)}$$

$\Rightarrow$ total energy (per unit volume)

$$\frac{E}{V} = \int_0^\infty \rho(\omega)\, d\omega = \frac{1}{\pi^2 c^3} \int_0^\infty \frac{\hbar \omega^3\, d\omega}{e^{\hbar\omega/k_B T} - 1} =$$

$$= \begin{pmatrix} x = \dfrac{\hbar\omega}{k_B T} \\ \omega = \dfrac{k_B T}{\hbar} x \end{pmatrix} = \frac{\hbar}{\pi^2 c^3} \left(\frac{k_B T}{\hbar}\right)^4 \int_0^\infty \frac{x^3\, dx}{e^x - 1}$$

From (5.110): $\displaystyle\int_0^\infty \frac{x^{s-1}\, dx}{e^x - 1} = \Gamma(s)\, \zeta(s)$

$$\Rightarrow \int_0^\infty \frac{x^3\, dx}{e^x - 1} = \Gamma(4)\, \zeta(4) = 3!\, \zeta(4) = 6 \cdot \frac{\pi^4}{90} = \frac{\pi^4}{15}$$

$$\Rightarrow \frac{E}{V} = \frac{\hbar}{\pi^2 c^3} \left(\frac{k_B T}{\hbar}\right)^4 \frac{\pi^4}{15} = \frac{\pi^2 k_B^4}{15 \hbar^3 c^3} T^4$$

P5.37 I'll do the problem in a slightly ~~differ~~ different from the suggested in the book way.

(a) Distinguishable particles in 3d harmonic potential

$$E_n = \hbar\omega\left(n_1+\tfrac{1}{2}+n_2+\tfrac{1}{2}+n_3+\tfrac{1}{2}\right) = \hbar\omega\left(n+\tfrac{3}{2}\right)$$

where $n_{1,2,3}$ describe 1d oscillator in x, y, z directions, correspondingly (re-visit Ch 10.5 of Townsend) eq.(10.81) + (10.82)

$$n_1, n_2, n_3 = 0, 1, 2 \ldots$$

The degeneracy $\underline{\underline{d}}(n) = \frac{1}{2}(n+1)(n+2)$ appears because, for fixed n, there are many ($\underline{\underline{d}}$) ways to satisfy

$$n = n_1 + n_2 + n_3.$$

We need to find

(1) $$N = \sum_n N_n = \sum_n d_n e^{-(\alpha+\beta E_n)}$$

(2) $$E = \sum_n E_n N_n = \sum_n d_n E_n e^{-(\alpha+\beta E_n)}$$

But $$N = \sum_n N_n = \sum_{n_1 n_2 n_3 = 0}^{\infty} e^{-\alpha-\beta\hbar\omega\left(n_1+n_2+n_3+\frac{3}{2}\right)} =$$

$$= e^{-\alpha-\beta\hbar\omega\cdot\frac{3}{2}} \sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty}\sum_{n_3=0}^{\infty} e^{-\beta\hbar\omega n_1} e^{-\beta\hbar\omega n_2} e^{-\beta\hbar\omega n_3}$$

$$= e^{-\alpha-\beta\hbar\omega\cdot\frac{3}{2}} \left(\sum_{n_1=0}^{\infty} e^{-\beta\hbar\omega n_1}\right)^3$$

Now $$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \Rightarrow \sum_{n=0}^{\infty} e^{-\beta\hbar\omega n} = \frac{1}{1-e^{-\beta\hbar\omega}}$$

(so that $e^{-\beta\hbar\omega} = x$)

Thus $$N = \frac{e^{-\alpha - \frac{3}{2}\beta\hbar\omega}}{\left(1-e^{-\beta\hbar\omega}\right)^3} = e^{-\alpha}\frac{1}{8\left(\sinh\frac{\beta\hbar\omega}{2}\right)^3}$$

Next $$E = \sum_n E_n N_n = \sum_{n_1 n_2 n_3} \hbar\omega\left(n_1+n_2+n_3\right) e^{-\alpha-\beta\hbar\omega\left(n_1+n_2+n_3+\frac{3}{2}\right)}$$

$$= e^{-\alpha}\sum_{n_1 n_2 n_3} \hbar\omega\left(n_1+n_2+n_3+\frac{3}{2}\right) e^{-\beta\hbar\omega\left(n_1+n_2+n_3+\frac{3}{2}\right)} =$$

$$= e^{-\alpha}\sum_{n_1 n_2 n_3} -\frac{d}{d\beta} e^{-\beta\hbar\omega\left(n_1+n_2+n_3+\frac{3}{2}\right)}$$

But, from the previous page,

$$\sum_{n_1 n_2 n_3} e^{-\beta\hbar\omega\left(n_1+n_2+n_3+\frac{3}{2}\right)} = \frac{e^{-\beta\hbar\omega\cdot\frac{3}{2}}}{\left(1-e^{-\beta\hbar\omega}\right)^3} =$$

$$= \left(\frac{e^{-\beta\hbar\omega/2}}{1-e^{-\beta\hbar\omega}}\right)^3 = \frac{1}{\left(e^{\beta\hbar\omega/2}-e^{-\beta\hbar\omega/2}\right)^3} = \frac{1}{\left(2\sinh\frac{\beta\hbar\omega}{2}\right)^3}$$

$$\Rightarrow E = e^{-\alpha}\left(-\frac{d}{d\beta}\frac{1}{8\left(\sinh\frac{\beta\hbar\omega}{2}\right)^3}\right) =$$

$$= -e^{-\alpha}\,-\frac{3}{8}\,\frac{\cosh\frac{\beta\hbar\omega}{2}}{\left(\sinh\frac{\beta\hbar\omega}{2}\right)^4}\cdot\frac{\hbar\omega}{2} =$$

$$= e^{-\alpha}\,\frac{3}{16}\hbar\omega\,\frac{\cosh\frac{\beta\hbar\omega}{2}}{\left(\sinh\frac{\beta\hbar\omega}{2}\right)^4}$$

But $e^{-\alpha} = 8N\left(\sinh\frac{\beta\hbar\omega}{2}\right)^3$

$$\Rightarrow E = 8N\left(\sinh\frac{\beta\hbar\omega}{2}\right)^3 \frac{3\hbar\omega}{16} \frac{\cosh\frac{\beta\hbar\omega}{2}}{\left(\sinh\frac{\beta\hbar\omega}{2}\right)^4}$$

$$= N\frac{3\hbar\omega}{2}\frac{\cosh\frac{\beta\hbar\omega}{2}}{\sinh\frac{\beta\hbar\omega}{2}} = N\frac{3\hbar\omega}{2}\frac{e^{\beta\hbar\omega/2}+e^{-\beta\hbar\omega/2}}{e^{\beta\hbar\omega/2}-e^{-\beta\hbar\omega/2}} =$$

$$= N\frac{3\hbar\omega}{2}\frac{1+e^{-\beta\hbar\omega}}{1-e^{-\beta\hbar\omega}}$$

(b) let now $k_BT \ll \hbar\omega$: this is essentially $T=0$ limit

$$\beta = \frac{1}{k_BT} \Rightarrow \frac{\hbar\omega}{k_BT} \gg 1 \Rightarrow e^{-\frac{\hbar\omega}{k_BT}} \ll 1$$

$$E = \frac{3}{2}N\hbar\omega\frac{1+e^{-\hbar\omega/k_BT}}{1-e^{-\hbar\omega/k_BT}} \approx \frac{3}{2}N\hbar\omega\left(1+e^{-\frac{2\hbar\omega}{k_BT}}\right) = \frac{3}{2}N\hbar\omega$$

Thus $E = N\cdot\underbrace{\frac{3}{2}\hbar\omega}$ → the lowest energy state of 3D oscillator, $n=0$.

[All ~~bosons~~ particles are in the ground state $n=0$

(c) $k_BT \gg \hbar\omega \Rightarrow \frac{\hbar\omega}{k_BT} \ll 1 \Rightarrow e^{-\frac{\hbar\omega}{k_BT}} = 1-\frac{\hbar\omega}{k_BT}$

$$\Rightarrow E = N\frac{3}{2}\hbar\omega\frac{1+1-\frac{\hbar\omega}{k_BT}}{1-\left(1-\frac{\hbar\omega}{k_BT}\right)} = N\frac{3}{2}\hbar\omega\frac{2}{\frac{\hbar\omega}{k_BT}} =$$

$$= N\cdot 3k_BT = N\cdot 6\cdot\left(\frac{1}{2}k_BT\right)$$

N particles, 6 degrees of freedom for each, $\frac{1}{2}k_BT$ for every degree.